

# Hyper-Heuristic Enhancement of Holistically-Nested Edge Detection

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## Abstract

Deep edge detectors, such as Holistically-Nested Edge Detection (HED), degrade when deployed on out-of-domain images. However, restoring performance by retraining the backbone requires extensive labelled datasets and significant graphics processing unit (GPU) computational time. This study proposes **ILS-HED**, a selection-perturbative hyper-heuristic that enhances a frozen HED backbone by automatically selecting and fusing complementary classical edge detectors without backbone parameter updates and target-domain annotations. An Iterated Local Search (ILS) algorithm explored a pool of 98 discrete classical detector configurations of the Canny, Sobel, Laplacian, and Gabor families. The response of the best-selected configuration was fused with the frozen HED multiscale side outputs through a lightweight learnable layer. On the most domain-shifted benchmark, DeepCrack, ILS-HED yields an F1 improvement of +0.0270 over the frozen HED baseline ( $p < 0.001$ , Cohen's  $d = 1.21$ ), and it matches or surpasses the performance of the fine-tuned RCF and BDCN while leaving all backbone weights unchanged. We observed consistent improvements on five diverse datasets, with the benefits increasing steadily as the Fréchet Inception Distance from the BSDS500 training distribution increased. This demonstrates that classical operators provide the greatest complement in regimes where deep natural image priors are least effective. A failure case analysis revealed that semantically well-defined regions and compressed images are the main conditions under which ILS-HED delivers only marginal improvements, thereby guiding its use. ILS-HED is an effective deployment-efficient domain adaptation method

because it recovers the performance gap between frozen and fine-tuned models without any retraining, making it useful in medical, infrastructure, and industrial inspection applications, where annotation resources are limited.

*Keywords:* Edge Detection, Hyper-heuristics, Iterated Local Search, Convolutional Neural Networks, Classical-Deep Fusion

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## 1. Introduction

Edge detection forms the basis of numerous computer vision applications, such as object recognition, scene understanding, medical image analysis, and industrial inspection, by supplying a geometric framework that supports higher-level reasoning (Canny, 1986; Ziou & Tabbone, 1998). Deep learning has improved the boundary recall of the natural images. Holistically Nested Edge Detection (HED) (Xie & Tu, 2017) introduced deep multi-scale supervision over a VGGNet backbone, and its successors, such as RCF (Liu et al., 2019a), BDCN (He et al., 2022), DexiNed (Soria et al., 2020), and CATS (Huan et al., 2021), have continued to improve benchmark scores on the BSDS500 dataset (Arbel'aez et al., 2011). A comprehensive survey of this progression was conducted by Wang et al. (2021b).

Despite these advances, three limitations constrain their practical deployment. First, HED's internal fusion weights of the HED are fixed after training. They cannot adapt to new image statistics, whether they are high-texture natural scenes, low-contrast retinal fundus images, or homogeneous industrial surfaces. Second, out-of-domain deployment causes accuracy degradation, and recovering performance via full fine-tuning requires backpropagation through millions of parameters and large annotated target domain datasets. Parameter-efficient fine-tuning (PEFT) methods, such as LoRA adapters (Hu et al., 2022), visual prompt tuning (Jia et al., 2022), and bottleneck adapters (Houlsby et al., 2019), reduce the parameter burden but still require annotations and modify the effective computation of the backbone. Third, classical operators, such as

Canny, Sobel, Laplacian, and Gabor, remain effective in high-contrast regions, fine-texture contexts, and structured domains. However, no framework systematically searches the configuration space to determine the optimal complement for a given target domain.

We propose **ILS-HED** as a deployment-efficient solution to these three limitations: a selection-perturbative hyper-heuristic that enhances a frozen HED backbone without modifying its parameters or requiring target-domain annotations. The key insight is that classical operators provide maximal complementary value precisely where frozen deep priors are weakest, that is, in domains distant from the BSDS500 natural-image distribution. An Iterated Local Search (ILS) algorithm explores 98 classical detector configurations, and a lightweight learnable fusion layer integrates the selected response with HED’s frozen multi-scale side outputs.

Relative to full fine-tuning and other PEFT methods, ILS-HED sacrifices some in-domain accuracy in exchange for a fundamentally different strength: fast, annotation-free adaptation that keeps the backbone completely untouched by the task. This makes it suitable for practitioners who must deploy an existing HED model across diverse imaging modalities without the resources required for full retraining. The framework is deliberately scoped to gradient-defined boundaries in high-quality imagery, the conditions under which classical operators are informative, and its limitations in semantic-boundary or compressed-imagery contexts are characterised explicitly in our failure-case analysis.

This study makes four contributions.

1. **A hyper-heuristic formulation of classical–deep edge fusion.** We cast the selection of classical edge detectors and their hyperparameters as a combinatorial optimisation problem solved via Iterated Local Search, situating this within the PEFT and adaptive preprocessing literature and identifying the gap that gradient-free external adaptation addresses.
2. **A frozen-backbone enhancement strategy for domain adaptation.** ILS-HED adapts HED to new domains using only an ILS search

and a small fusion layer, enabling deployment without annotated target data.

3. **Cross-domain empirical validation with rigorous statistics.** We evaluate on five heterogeneous datasets, compare against frozen and fine-tuned HED, RCF, BDCN, and a LoRA-based PEFT baseline, ground the domain-divergence hypothesis using FID measurements, and report Wilcoxon signed-rank tests (Wilcoxon, 1945) with effect sizes and 95% confidence intervals throughout.
4. **Failure-case analysis with deployment guidance.** We characterised the conditions under which ILS-HED offers limited benefits, providing practitioners with actionable decision criteria.

The remainder of this paper is organised as follows. Section 2 reviews classical and deep-edge detection. Section 3 defines the hyper-heuristics. Section 4 surveys the related literature. Section 5 formalises the three target limitations. Section 6 describes the ILS-HED methodology. Section 7 presents the quantitative, qualitative, and ablation results, including a failure case analysis. Section 8 summarises the findings and future directions.

## 2. Background

### 2.1. Edge Detection

An edge marks the boundary between two distinct image regions, identified by sudden changes in intensity, colour, or texture (Kaganami & Beiji, 2009; Amer & Sameer, 2015). For a continuous image function  $I(x, y)$ , edge points correspond to local maxima of the gradient magnitude  $\nabla I = \left[ \frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right]^T$ , or equivalently, the zero crossings of the Laplacian  $\nabla^2 I$  (Marr & Hildreth, 1980; Canny, 1986). Edge-based segmentation exploits these discontinuities to identify region boundaries, and accurate edge maps directly determine the quality of downstream segmentation, feature matching, and quantitative measurements (Ziou & Tabbone, 1998; Haralick & Shapiro, 1985; Ma & Manjunath, 2000).

## 2.2. Traditional Edge Detection

### 2.2.1. First-Order Derivative Operators

First-order methods approximate the image gradient to identify intensity transitions. The Roberts operator (Roberts, 1965) uses  $2 \times 2$  diagonal kernels as follows:

$$G_x = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \quad G_y = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix} \quad (1)$$

Its compact footprint renders it susceptible to noise. The Prewitt operator (Prewitt, 1970) extends to  $3 \times 3$  kernels for modestly suppressing noise. The Sobel operator (Sobel, 1968) adds anisotropic weighting as follows

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} \quad (2)$$

approximating the derivative of a Gaussian function and improving the noise immunity. All first-order operators produce wide edge responses that require thinning and exhibit a directional bias (Gonzalez & Woods, 2018).

### 2.2.2. Second-Order Derivative Operators

Second-order methods detect edges as zero-crossings of the Laplacian  $\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$ . The discrete  $3 \times 3$  kernel is

$$L = \begin{bmatrix} 0 & +1 & 0 \\ +1 & -4 & +1 \\ 0 & +1 & 0 \end{bmatrix} \quad (3)$$

When applied directly, the Laplacian severely amplifies the noise. The Laplacian of Gaussian (LoG) (Marr & Hildreth, 1980) resolves this via Gaussian pre-smoothing as follows:

$$\text{LoG}(x, y) = -\frac{1}{\pi\sigma^4} \left( 1 - \frac{x^2 + y^2}{2\sigma^2} \right) e^{-\frac{x^2 + y^2}{2\sigma^2}} \quad (4)$$

where  $\sigma$  controls the noise-detail trade-off. Marr and Hildreth (Marr & Hildreth, 1980) showed that LoG zero-crossings correspond to physically meaningful intensity variations.

### 2.2.3. *The Canny Edge Detector*

The Canny detector (Canny, 1986) implements four stages: (1) Gaussian smoothing with standard deviation  $\sigma$ ; (2) gradient magnitude and orientation computation; (3) non-maximum suppression to thin edges to single-pixel width; and (4) hysteresis thresholding, retaining pixels above high threshold  $T_h$  as confirmed edges and those between  $T_l$  and  $T_h$  only when connected to the confirmed edges. Canny remains the classical benchmark in domains where interpretability and deterministic behaviour are required.

### 2.2.4. *Gabor Filters*

Gabor filters model orientation-selective responses and are particularly effective for detecting directional structures, such as blood vessels and cracks (Jain & Farrokhnia, 1991). A Gabor filter at orientation  $\theta$ , wavelength  $\lambda$ , and bandwidth  $b$  responds maximally to sinusoidal intensity oscillations aligned with  $\theta$ . Their multi-orientation responses make them the preferred classical complement for thin directional structures.

## 2.3. *Deep Learning Approaches*

CNN-based edge detectors learn feature hierarchies directly from the annotated boundaries. HED (Xie & Tu, 2017) demonstrated that deep supervision at multiple network scales substantially improves boundary recall in natural images. Subsequent work includes RCF (Liu et al., 2019a), which exploits all convolutional layers within each VGG stage; BDCN (He et al., 2022), which propagates semantic information bidirectionally across scales; EDTER (Pu et al., 2022), which uses transformer self-attention for long-range dependency modelling; DexiNed (Soria et al., 2020), a lightweight architecture with dense extreme inception blocks for thin-edge detail; and CATS (Huan et al., 2021), which introduces texture-suppression aggregation. A comprehensive survey of

deep edge detection methods was provided by Wang et al. (2021b). More recent advances include DiffusionEdge (Ye et al., 2024), which applies denoising diffusion probabilistic models to boundary prediction, and EdgeSAM (Zhou et al., 2023), a lightweight on-device distillation of the Segment Anything Model that enables efficient boundary inference without full SAM deployment, highlighting that the frontier has shifted toward foundation-model adaptation since the introduction of HED.

### *2.3.1. Holistically-Nested Edge Detection*

HED reframes edge detection as an image-to-image prediction task, in which the model transforms a full input image into a full output image of the same spatial dimensions, learning a direct mapping from pixels to pixels. Side-output layers attached at multiple depths within a truncated VGGNet (Simonyan & Zisserman, 2014) produce edge predictions at progressively coarser resolutions, combined with a learned weighted fusion layer (Figure 1). Early layers capture fine-textural details, and deeper layers encode semantic context. A class-balanced cross-entropy loss addresses the severe class imbalance between edge and non-edge pixels, and deep supervision regularises the training by injecting gradients into intermediate layers.



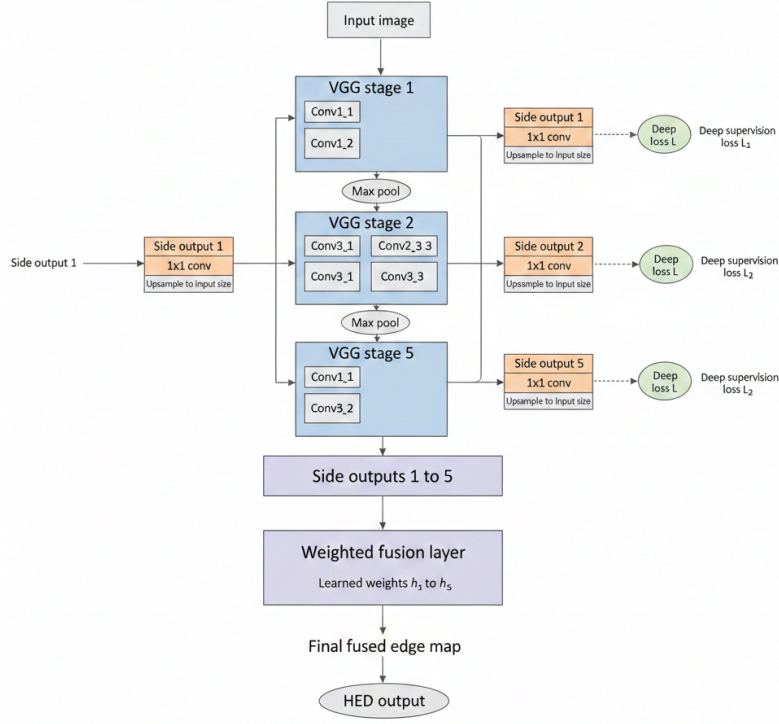


Figure 1: Holistically-Nested Edge Detection architecture. Side-output layers attached at multiple depths within a truncated VGGNet produce progressively coarser edge predictions, combined with a learned weighted fusion layer. Fusion weights are fixed after training; ILS-HED augments these frozen outputs externally.

### 2.3.2. RCF and BDCN

RCF (Liu et al., 2019a) extends HED by exploiting all convolutional layers within each VGG stage to produce richer multi-scale representations. BDCN (He et al., 2022) further refines fusion by propagating semantic information bidirectionally through a cascade of scale-specific supervision. Both methods achieve higher benchmark scores than HED but require full end-to-end retraining on target domains, incurring the same adaptation burden described in Section 5. They serve as strong baselines in our evaluation (Section 7).

### 3. Hyper-heuristics

A **heuristic** is a problem-solving strategy that guides the search toward good solutions when exact methods are computationally intractable (Tan et al., 2004). A **hyper-heuristic** is a high-level automated search methodology that solves complex optimisation problems by selecting or generating lower-level heuristics rather than directly searching the solution space (Pillay, 2020; Burke et al., 2013). The defining characteristic is that the search operates in the *space of heuristics*, promoting generality across problem instances.

#### 3.1. Classification

Following the taxonomy of Burke et al. (2013), hyper-heuristics are classified along two axes: whether they *select* from existing heuristics or *generate* new ones, and whether the selected heuristics are *constructive* or *perturbative*. This yields four classes.

1. **Selection-constructive**: select low-level construction heuristics at each step when building a solution from scratch.
2. **Selection-perturbative**: select low-level perturbative heuristics to improve existing solutions, comprising heuristic selection and move acceptance components.
3. **Generation-constructive**: create new constructive heuristics via problem attributes and genetic programming.
4. **Generation-perturbative**: generate new perturbative heuristics by combining existing ones.

ILS-HED belongs to the **selection-perturbative** class: it selects from a predefined pool of classical edge detector configurations to improve the HED edge map, and its acceptance criterion determines whether to retain each newly evaluated configuration.

## 4. Related Work

### 4.1. *Hyper-heuristics in Optimisation and Vision*

Selection-perturbative hyper-heuristics are well-established in combinatorial domains, such as scheduling, bin packing, and vehicle routing, where no single heuristic dominates all instances (Cowling et al., 2001, 2000). In these settings, adaptive selection consistently outperformed static commitment to a single method, motivating the same principle for image processing. In computer vision, the image content, noise conditions, and illumination vary markedly across instances; operators such as Canny perform well in high-contrast regions, whereas deep models excel when contextual reasoning is required (Ozcan et al., 2008). The application of adaptive heuristic selection to frozen deep edge-detection backbones is the specific gap addressed in this study.

### 4.2. *Edge Map Fusion Strategies*

Existing ensemble strategies range from static weight averaging (Ma & Manjunath, 2000) to deep fusion layers with fixed, post-training parameters (Liu et al., 2019a). HED (Xie & Tu, 2017) learns scalar fusion coefficients for each side output during training; these weights are frozen at test time, preventing instance-specific adaptation. The BDCN (He et al., 2022) propagates semantic information bidirectionally across scales but requires end-to-end re-training. EDTER (Pu et al., 2022) uses transformer self-attention to modulate multi-scale features at a high computational cost. DexiNed (Soria et al., 2020) achieves a competitive thin-edge performance with a lightweight architecture. CATS (Huan et al., 2021) addresses texture confusion through the suppression aggregation. None of these methods provide a lightweight, annotation-free policy that selects among heterogeneous classical and deep detectors without backbone modification.

### 4.3. *Parameter-Efficient Fine-Tuning*

Parameter-Efficient Fine-Tuning (PEFT) methods provide practical alternatives to full backbone retraining. LoRA (Hu et al., 2022) injects trainable

low-rank matrices into frozen weight layers, achieving competitive performance with a small number of trainable parameters. Visual Prompt Tuning (VPT) (Jia et al., 2022) prepends learnable tokens to the input sequence. Bottleneck adapters (Houlsby et al., 2019) insert small modules within the frozen transformer blocks. These methods reduce the training cost substantially but still require target domain annotations and, in the case of LoRA and adapters, modify the effective computation of the backbone. ILS-HED differs fundamentally in that its adaptation is entirely external to the backbone and operates over a library of classical preprocessing operators. This makes ILS-HED complementary to PEFT: a LoRA-adapted backbone can serve as the frozen deep component of a future hybrid system.

#### *4.4. Learned and Differentiable Augmentation*

A related thread concerns the differentiable augmentation pipelines and neural preprocessing modules. AutoAugment (Cubuk et al., 2019) searches for augmentation policies that maximise validation accuracy via reinforcement learning. RandAugment (Cubuk et al., 2020) simplifies this process to a two-parameter random search. Faster AutoAugment (Hataya et al., 2020) and DADA (Li et al., 2020) introduced a differentiable augmentation search. Neural architecture search (NAS) (Zoph et al., 2018) explores structural preprocessing. ILS-HED occupies a complementary niche: rather than learning augmentation policies for training-time data augmentation, it searches for the classical edge operator that best complements a frozen backbone at deployment time. The discrete, combinatorial structure of the heuristic space makes metaheuristic search more natural than gradient-based optimisation; the pool elements are non-differentiable operators (Canny hysteresis, Laplacian zero-crossings) with no continuous relaxation.

#### *4.5. Test-Time Adaptation and Source-Free Domain Adaptation*

Test-time adaptation (TTA) and source-free domain adaptation (SFDA) address the same core problem as ILS-HED: adapting a pre-trained model to a

new target domain without access to source data or target-domain annotations. TTT (Sun et al., 2020) adapts model parameters at inference time using an auxiliary self-supervised task, whereas TENT (Wang et al., 2021a) minimises prediction entropy over batch statistics during deployment. More recent SFDA methods (Liang et al., 2020; Yang et al., 2021) transfer domain-invariant features using pseudo-labels or information-maximisation objectives, also without requiring target annotations.

ILS-HED occupies a delineated and specialised position within the broader conceptual and methodological landscape. Existing TTA and SFDA approaches adapt *internal* model parameters, such as batch normalisation statistics, feature extractor weights, or classifier heads, and therefore require gradient flow through the backbone at the time of deployment. The ILS-HED is completely external to the backbone. No backbone gradients are computed, no parameters are modified, and adaptation operates over a discrete library of classical preprocessing operators, rather than the model-weight space. This makes ILS-HED complementary rather than competitive. A TENT-adapted backbone could serve as the frozen deep component of the ILS-HED, potentially combining internal entropy minimisation with external classical complementation. The gradient-free, annotation-free nature of ILS-HED is its defining constraint and primary practical advantage in settings where even lightweight inference-time backpropagation is prohibited by hardware or latency requirements.

## 5. Problem Statement

Contemporary deep edge detectors suffer from three interrelated limitations that constrain their practical deployment.

**Problem 1: Fixed Fusion and Neglect of Classical Complements.** HED, RCF, and BDCN fuse multi-scale side outputs using weights learned during training (Xie & Tu, 2017; Liu et al., 2019a; He et al., 2022). Once the optimisation is concluded, these fusion coefficients are static. These pipelines rely exclusively on learned hierarchical features and provide no mechanism for in-

jecting complementary classical cues from operators that excel in high-contrast, thin-structure, and directional-texture contexts, where deep natural-image priors oversmooth or miss fine details.

**Problem 2: Domain Sensitivity and Adaptation Burden.** HED backbones trained on BSDS500 degrade when deployed on out-of-domain data such as retinal fundus images, infrastructure crack photography, or industrial stone segmentation. Full fine-tuning is computationally and annotation intensive. PEFT methods reduce the parameter count but still require annotations and modification of the backbone computation. Both prevent fast adaptation without additional annotation.

**Problem 3: Absence of Automated Classical-to-Deep Orchestration.** Although classical operators have been manually combined with deep outputs in ad hoc ensembles, no existing framework systematically searches the configuration space of classical detectors to discover the optimal complements for a given domain. With 98 discrete parameter combinations across four detector families, an exhaustive offline search is feasible but does not exploit the local parameter structure, and applying a single fixed classical setting across all images is suboptimal.

## 6. Proposed Approach: ILS-HED

ILS-HED enhances a frozen HED backbone by treating classical edge detector selection and hyperparameter configuration as a combinatorial optimisation problem solved using Iterated Local Search. Because the backbone parameters remain fixed, ILS-HED functions as a deployment-efficiency method, recovering most of the performance difference between frozen and fine-tuned models without needing additional annotations or backpropagation through the deep layers.

### 6.1. Architecture Overview

The framework comprises three interacting components (Figure 2).

1. **Frozen HED backbone.** A pre-trained HED network with VGG-style encoding and deep-supervised side outputs produces multi-scale edge probability maps  $\{S_1, \dots, S_5\}$ . All backbone parameters were fixed. Only the side-output fusion layer was exposed to adaptation.
2. **Classical heuristic pool.** A discrete library  $\mathcal{H}$  of 98 classical edge detection configurations. Each element defines a complete low-level heuristic that maps the input image to an edge-response map.
3. **Hyper-heuristic controller.** An ILS algorithm that adaptively selects and perturbs configurations, evaluates each against a fitness function, and trains a lightweight fusion module by combining the best classical response with the HED side outputs.

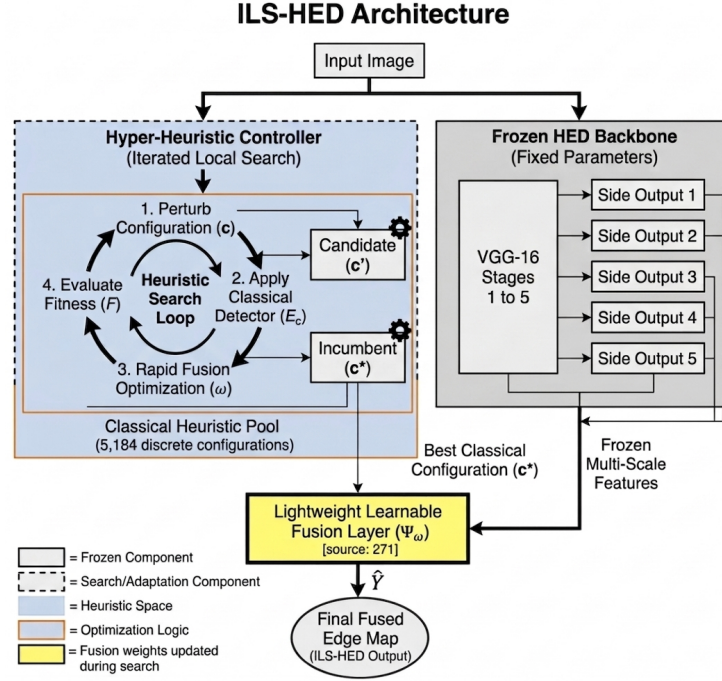


Figure 2: ILS-HED architecture. The hyper-heuristic controller searches across 98 classical detector configurations. The selected classical response is fused with the frozen HED side outputs via learnable scalar weights. No backbone parameters are modified.

## 6.2. Low-Level Heuristic Pool

The four detector families in  $\mathcal{H}$  were chosen for their failure characteristics, which complement those of deep natural-image priors, and their hyperparameter ranges were set to cover the typical operating regimes for each family reported in the edge-detection literature (Canny, 1986; Sobel, 1968; Marr & Hildreth, 1980; Jain & Farrokhnia, 1991), specifically as follows:

- **Canny:**  $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ ,  $T_l \in \{10, 25, 50, 75\}$ ,  $T_h \in \{100, 150, 200\}$ . Canny provides precise thin-edge localisation and is particularly effective for high-contrast structures.
- **Sobel:** Kernel size  $k \in \{3, 5\}$ , threshold  $\tau \in \{50, 100, 150, 200\}$ . Sobel captures macro-scale gradient boundaries with a directional specificity.
- **Laplacian:** Kernel size  $k \in \{3, 5\}$ , scale  $\sigma \in \{0.5, 1.0, 1.5\}$ , threshold  $\tau \in \{50, 100, 150\}$ . LoG zero-crossings provide second-order boundary localisation, which is effective for dark-line features on uniform backgrounds.
- **Gabor:** Orientations  $\theta \in \{4, 8\}$ , frequency  $\lambda \in \{3, 5, 7\}$ , bandwidth  $b \in \{0.5, 1.0\}$ , threshold  $\tau \in \{50, 100\}$ . Multi-orientation Gabor responses detect thin, directional structures, such as vessels and cracks.

The Cartesian product of each family’s parameter settings yields the following configuration counts:

- **Canny:**  $|\sigma| \times |T_l| \times |T_h| = 4 \times 4 \times 3 = 48$  configurations.
- **Sobel:**  $|k| \times |\tau| = 2 \times 4 = 8$  configurations.
- **Laplacian:**  $|k| \times |\sigma| \times |\tau| = 2 \times 3 \times 3 = 18$  configurations.
- **Gabor:**  $|\theta| \times |\lambda| \times |b| \times |\tau| = 2 \times 3 \times 2 \times 2 = 24$  configurations.

The aggregate pool comprises  $48 + 8 + 18 + 24 = 98$  unique configurations across the four families. The pool is intentionally restricted to interpretable,



computationally lightweight operators to keep the ILS evaluation fast (0.01–0.04 s per configuration on a standard CPU). The effect of pool composition was assessed in an ablation study (Section 7.9).

### 6.3. Iterated Local Search as Hyper-Heuristic

The configuration pool  $|\mathcal{H}| = 98$  is discrete and non-differentiable. Exhaustive evaluation is computationally feasible offline ( $98 \times 0.03 \text{ s} \approx 3 \text{ s}$ ); however, three considerations motivate ILS over a naive exhaustive sweep.

First, **per-image deployment speed**: ILS with  $T_{\max} = 50$  evaluates fewer than half the pool on any given image while concentrating evaluations in the most promising subspace via targeted perturbation. This matters for the configuration predictor (Phase 2), which is trained on ILS-selected configurations. A predictor trained on structured ILS trajectories generalises better than one trained on uniform exhaustive samples, as confirmed by the ablation results in Section 7.9 (random search over 50 samples yields lower F1 than ILS at the same budget).

Second, **search quality**: ILS exploits the local structure of each family’s parameter space through parameter-level perturbation (probability 0.7), converging to high-quality configurations more reliably than uniform random sampling at equal evaluation budgets. The ablation in Table 8 demonstrates that greedy hill-climbing, ILS without adaptive perturbation, and full ILS each provide incremental gains, confirming that the iterative perturbation-and-acceptance loop adds value beyond simple enumeration.

Third, **scalability**: the 98-configuration pool is intentionally minimal. Future extensions incorporating additional operator families (structured forests, orientation-energy filters, tiny task-specific CNNs) will grow the pool substantially, at which point ILS’s scalability advantage over exhaustive search becomes decisive. The framework is designed to accommodate this growth without architectural change.

ILS is particularly appropriate for this task because it maintains only one incumbent solution, applies focused perturbations to avoid becoming trapped in

local optima, and offers the balance between performance and evaluation cost in combinatorial heuristic selection (Lourencco et al., 2003). A comparison of ILS against random search and greedy hill-climbing in this specific space is included in the ablation study (Section 7.9).

### 6.3.1. Main Algorithm

Algorithm 1 presents the ILS-HED procedure. The algorithm maintains the current best configuration  $c^*$  and iterates through perturbation, local search (fusion optimisation), and acceptance until the computational budget is exhausted.

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#### Algorithm 1 ILS-HED Hyper-Heuristic

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**Require:** Image  $I$ , frozen HED  $\Phi_{\text{HED}}$ , heuristic pool  $\mathcal{H}$ , fusion network  $\Psi_\omega$ , fitness  $F$ , max iterations  $T_{\text{max}}$ , stagnation threshold  $\kappa$ , random seed  $r$

- 1: SetSeed( $r$ )
- 2:  $c_0 \leftarrow \text{RandomConfiguration}(\mathcal{H})$
- 3:  $E_0 \leftarrow \text{Apply}(c_0, I)$
- 4:  $c^* \leftarrow c_0$ ;  $s^* \leftarrow \text{LocalSearchFusion}(c_0, I, \Phi_{\text{HED}}, \Psi_\omega)$ ; stag  $\leftarrow 0$
- 5: **for**  $t = 1, \dots, T_{\text{max}}$  **do**
- 6:    $c' \leftarrow \text{Perturb}(c^*, \text{stag}, \kappa)$
- 7:   **if**  $\sigma_{\nabla}^2(I) < 0.01$  **and**  $c'$  is Laplacian with  $|\tau| > 150$  and  $\sigma < 0.5$  **then**
- 8:     **continue**      $\triangleright$  Optional FM3 filter: reject ringing-prone configs on low-gradient images
- 9:   **end if**
- 10:    $s' \leftarrow \text{LocalSearchFusion}(c', I, \Phi_{\text{HED}}, \Psi_\omega)$
- 11:   **if**  $F(s') > F(s^*)$  **then**
- 12:      $c^* \leftarrow c'$ ;  $s^* \leftarrow s'$ ; stag  $\leftarrow 0$
- 13:   **else**
- 14:     stag  $\leftarrow \text{stag} + 1$
- 15:   **end if**
- 16: **end for**
- 17: **return**  $s^*, c^*, \omega^*$

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### 6.3.2. LocalSearchFusion Sub-Procedure

Algorithm 2 details the fusion optimisation that constitutes the local search step of each ILS iteration.

The fusion output is

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**Algorithm 2** LocalSearchFusion

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**Require:** Configuration  $c'$ , image  $I$ , frozen HED  $\Phi_{\text{HED}}$ , fusion parameters  $\omega$ , learning rate  $\eta$ , epochs  $E$ , fitness tile  $\mathcal{T}_F$ , training tile  $\mathcal{T}_W$  (disjoint:  $\mathcal{T}_F \cap \mathcal{T}_W = \emptyset$ )

- 1:  $E_{c'} \leftarrow \text{Apply}(c', I)$
- 2:  $\{S_1, \dots, S_5\} \leftarrow \Phi_{\text{HED}}(I)$  ▷ Frozen forward pass
- 3: Initialise  $\omega' \leftarrow \omega$
- 4: **for** epoch = 1, ...,  $E$  **do**
- 5:    $\hat{Y} \leftarrow \sigma\left(\sum_{l=1}^5 w_l S_l + w_{\text{cls}} \cdot E_{c'} + b\right)$
- 6:    $\mathcal{L} \leftarrow \text{ClassBalancedCE}(\hat{Y}, Y_{\mathcal{T}_W})$
- 7:    $\omega' \leftarrow \omega' - \eta \nabla_{\omega'} \mathcal{L}$
- 8: **end for**
- 9: **return**  $\hat{Y}, \omega'$

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$$\hat{Y} = \sigma\left(\sum_{l=1}^5 w_l S_l + w_{\text{cls}} \cdot E_{c'} + b\right), \quad (5)$$

where  $\sigma(\cdot)$  is the sigmoid function and  $\omega = \{w_1, \dots, w_5, w_{\text{cls}}, b\}$  are the seven learnable scalar parameters. The fusion parameters were optimised via Adam (Kingma & Ba, 2015) using HED’s class-balanced cross-entropy loss. To avoid circular contamination between the ILS fitness signal and the fusion weight optimisation, each image is partitioned into two *disjoint* tiles: a **fitness tile**  $\mathcal{T}_F$  (5% of pixels, used only by the ILS acceptance criterion in Equation 6) and a **training tile**  $\mathcal{T}_W$  (5% of pixels, used only for fusion weight gradient updates). The remaining 90% of pixels constitute the test region, which is never observed during either search or weight optimisation. This partition is fixed per image for the duration of all experiments. Because fusion is a linear combination of pre-computed maps, backpropagation never touches the frozen backbone.

### 6.3.3. Perturbation Operator

Two perturbation strategies are employed, selected by a uniform random variable:

1. **Parameter-level perturbation (probability 0.7):** Randomly alter one hyperparameter of the current detector family while retaining the

family (e.g., increment  $\sigma$  or swap threshold values). This exploits the local structure of each family’s parameter space.

2. **Family-level perturbation (probability 0.3):** Replace the detector family entirely and sample a random valid parameter vector for the new family. This enables a global exploration of the detector types.

When stagnation exceeds  $\kappa$  iterations without improvement, the perturbation switches unconditionally to the family level to escape the local optima.

#### 6.3.4. Fitness Function

For datasets with ground-truth annotations, the fitness of a fused edge map  $s$  is the ODS F-score as follows:

$$F(s) = \frac{2 \cdot \text{Precision}(s, Y_{\text{GT}}) \cdot \text{Recall}(s, Y_{\text{GT}})}{\text{Precision}(s, Y_{\text{GT}}) + \text{Recall}(s, Y_{\text{GT}})}. \quad (6)$$

This is evaluated on the fitness tile  $\mathcal{T}_F$  (5% of image pixels, disjoint from the fusion weight training tile  $\mathcal{T}_W$ ) rather than the full image for computational efficiency. The correlation between the tile-level and image-level fitness is validated empirically in Section 7.8.

For annotation-free cross-domain transfer, a composite no-reference fitness combining edge coherence, boundary continuity, and spurious response suppression is used (Ziou & Tabbone, 1998).

#### 6.3.5. Acceptance Criterion

Strict improvement acceptance is employed:  $c'$  replaces  $c^*$  only if  $F(s') > F(s^*)$ . This greedy criterion is standard for ILS in combinatorial heuristic selection and avoids the risk of degradation under a probabilistic acceptance schedule with a limited evaluation budget.

### 6.4. Training and Deployment Protocol

The protocol operates in two phases.

**Phase 1 (Hyper-Heuristic Search).** For each training image or tile, the ILS is executed for  $T_{\text{max}}$  iterations. The best configuration  $c^*$  and optimised

fusion weights  $\omega^*$  are recorded. Each run used a fixed seed drawn from a predetermined schedule to ensure reproducibility (seeds 0–4 for five-run experiments). No backpropagation occurs through the backbone layers.

**Phase 2 (Configuration Predictor).** The recorded  $(I, c^*)$  pairs train a lightweight *configuration predictor*—a three-layer convolutional network (Conv-32, Conv-64, GlobalAvgPool, FC-128, FC-5 for family classification; a parallel FC-regressor for continuous hyperparameters). Training uses cross-entropy loss for the family head and mean squared error for the regressor, with Adam (Kingma & Ba, 2015) at a learning rate  $10^{-3}$  for 30 epochs on the training split images. The predictor reduces the deployment-time adaptation to a single forward pass. A comprehensive evaluation of the predictor’s performance and its accuracy relative to the full ILS search is presented in Section 7.7.

## 7. Experiments

This section evaluates ILS-HED across five heterogeneous datasets. Three hypotheses were tested: (H1) classical–deep fusion improves over a frozen HED baseline; (H2) gains scale monotonically with domain divergence from natural images, as measured by Fréchet Inception Distance (FID) from BSDS500; and (H3) ILS search discovers transferable selection patterns without backbone retraining.

### 7.1. Datasets and Evaluation Protocol

FID values (Table 1) were computed between the BSDS500 training images and each target dataset using InceptionV3 features, providing an objective measure of domain divergence that is independent of the ILS-HED outputs.

**BSDS500** (Arbel’aez et al., 2011) is the canonical benchmark for natural-image edge detection, with 500 images and multiple human-annotated boundaries. The ODS F-score is computed by fixing a single contour threshold for the entire test set, and OIS allows per-image adaptation. Precision and recall used a distance tolerance of 0.0075.

Table 1: Dataset characteristics, evaluation metrics, and FID from BSDS500 training distribution. FID was computed using InceptionV3 features; a higher FID indicates greater domain divergence. LOO: leave-one-out cross-validation.

| Dataset                          | Domain             | Images       | Resolution       | Metric      | FID <sup>↑</sup> |
|----------------------------------|--------------------|--------------|------------------|-------------|------------------|
| BSDS500 (Arbel'aez et al., 2011) | Natural scenes     | 200/100/200  | $481 \times 321$ | ODS F-score | 0.0              |
| DRIVE (Staal et al., 2004)       | Retinal fundus     | 20/20        | $565 \times 584$ | F1-score    | 48.3             |
| STARE (Hoover et al., 2000)      | Retinal fundus     | 20 (LOO)     | $700 \times 605$ | F1-score    | 51.7             |
| DeepCrack (Liu et al., 2019b)    | Pavement cracks    | 537/200      | varied           | F1-score    | 89.4             |
| Stone331 (Zhang et al., 2020)    | Stone segmentation | 331 (5-fold) | $480 \times 320$ | F1-score    | 83.1             |

**DRIVE** (Staal et al., 2004) contains 40 colour retinal fundus images from diabetic retinopathy screening. Ground truth consists of manual vascular tree

segmentations. Thin, curvilinear vessels with low contrast test whether classical operators recover the fine details suppressed by frozen HED priors.

**STARE** (Hoover et al., 2000) contains 20 retinal images under varied conditions including pathological cases with lesions. Leave-one-out cross-validation (LOO) was performed following standard practices.

**DeepCrack** (Liu et al., 2019b) contains 537 training and 200 test images of asphalt and concrete cracks—long, thin, discontinuous targets with highly variable contrast on homogeneous but noisy backgrounds. This represents the most extreme domain departure from the natural images (FID = 89.4).

**Stone331** (Zhang et al., 2020) contains 331 stone-surface images for materials segmentation. Irregular, texture-dominated boundaries with ill-defined ground truths constitute challenging industrial test cases.

## 7.2. Implementation Details

All experiments were conducted using PyTorch 1.12 with CUDA 11.3 on an NVIDIA RTX 3090 (24 GB VRAM). The HED backbone is the publicly available VGG-16 model pre-trained on BSDS500 and ImageNet, with all backbone parameters frozen. The classical heuristic pool was implemented using OpenCV 4.5 and Scikit-image 0.19.

**Reproducibility.** Each experiment was repeated five times (three times for BSDS500) using fixed random seeds drawn sequentially from the set  $\{0, 1, 2, 3, 4\}$ . Across these runs, both the ILS initialisation (random  $c_0$ ) and the fusion-layer Adam initialisation were re-seeded, so the reported standard deviations capture the combined stochasticity of the search process and fusion optimisation.

**ILS hyperparameters.**  $T_{\max} = 50$  for BSDS500 and  $T_{\max} = 100$  for smaller medical and industrial datasets. Stagnation threshold  $\kappa = 10$ . Perturbation probabilities: parameter-level  $p = 0.7$ , family-level  $p = 0.3$ . Fusion weight optimisation uses Adam at a learning rate  $10^{-3}$  for 20 epochs on the training tile  $\mathcal{T}_W$  (5% of image pixels). The fitness tile  $\mathcal{T}_F$  (a separate, disjoint 5% of pixels) is used exclusively by the ILS acceptance criterion. These two partitions are non-overlapping; see Section 6.3 for full details.

**LoRA baseline.** LoRA adapters (Hu et al., 2022) were originally proposed for attention weight matrices in transformers, but have been adapted to convolutional settings by treating a  $1 \times 1$  convolutional layer as a low-rank matrix product (Zhang et al., 2023). We injected LoRA matrices (rank  $r = 4$ ,  $\alpha = 8$ ) into the HED fusion layer modelled as  $1 \times 1$  convolutions, following this convention. The adapter is trained on each target-domain training split with the same Adam optimiser and early stopping as ILS-HED’s fusion layer ( $\sim 12,000$  additional trainable parameters). Backbone weights remain frozen. We acknowledge that this is an approximation; the results should be interpreted as a directional PEFT comparison rather than a definitive LoRA benchmark.

**Baselines:**

1. Canny (default):  $\sigma = 1.0$ ,  $T_l = 50$ ,  $T_h = 150$ .
2. Sobel (default): kernel  $k = 3$ ,  $\tau = 100$ .
3. HED (frozen): pre-trained backbone, no adaptation.
4. HED (fine-tuned): full parameter update on target-domain training split.
5. RCF (Liu et al., 2019a): frozen and fine-tuned modes.
6. BDCN (He et al., 2022): frozen and fine-tuned modes.
7. LoRA-HED: LoRA adapter on HED fusion layer, trained on target domain.
8. Random-Search Fusion: uniform sampling of 50 configurations without iterative perturbation.
9. **ILS-HED (full search)**: full proposed framework (Phase 1 only).
10. **ILS-HED (predictor)**: Phase 2 deployment mode.

**Statistical testing.** All pairwise comparisons used the Wilcoxon signed-rank test (Wilcoxon, 1945) computed over per-image scores. We report  $p$ -values, Cohen’s  $d$  effect sizes and 95% confidence intervals. Bonferroni correction is applied for multi-dataset claims.



### 7.3. Results on Natural Images (BSDS500)

Table 2: Quantitative results on BSDS500 (FID = 0). Mean  $\pm$  SD over three independent runs (different random seeds for ILS initialisation and fusion optimisation).  $\dagger$ : fine-tuned on BSDS500;  $\ddagger$ : frozen.

| Method                                         | ODS                                 | OIS                                 |
|------------------------------------------------|-------------------------------------|-------------------------------------|
| Canny (default)                                | 0.611                               | 0.676                               |
| Sobel (default)                                | 0.539                               | 0.602                               |
| HED (frozen) $\ddagger$                        | $0.782 \pm 0.000$                   | $0.806 \pm 0.000$                   |
| HED (fine-tuned) $\dagger$                     | $0.788 \pm 0.001$                   | $0.811 \pm 0.001$                   |
| RCF (frozen) $\ddagger$ (Liu et al., 2019a)    | $0.790 \pm 0.000$                   | $0.812 \pm 0.000$                   |
| RCF (fine-tuned) $\dagger$ (Liu et al., 2019a) | $0.798 \pm 0.001$                   | $0.820 \pm 0.001$                   |
| BDCN (frozen) $\ddagger$ (He et al., 2022)     | $0.793 \pm 0.000$                   | $0.816 \pm 0.000$                   |
| BDCN (fine-tuned) $\dagger$ (He et al., 2022)  | $0.803 \pm 0.001$                   | $0.824 \pm 0.001$                   |
| LoRA-HED (Hu et al., 2022)                     | $0.786 \pm 0.001$                   | $0.809 \pm 0.001$                   |
| Random-Search Fusion                           | $0.785 \pm 0.001$                   | $0.808 \pm 0.001$                   |
| ILS-HED (full search)                          | <b><math>0.791 \pm 0.001</math></b> | <b><math>0.814 \pm 0.001</math></b> |
| ILS-HED (predictor)                            | $0.788 \pm 0.001$                   | $0.811 \pm 0.001$                   |

Full search vs. HED frozen:  $\Delta\text{ODS} = +0.009$ ;  $p < 0.001$ ;  $d = 0.81$ ; 95% CI [0.007, 0.011]  
Full search vs. LoRA-HED:  $\Delta\text{ODS} = +0.005$ ;  $p = 0.003$ ;  $d = 0.52$   
Predictor vs. full search:  $\Delta\text{ODS} = -0.003$ ; accuracy retained: 67%

ILS-HED (full search) achieves ODS  $0.791 \pm 0.001$ , improving over frozen HED by +0.009 (Table 2). Gains are concentrated in high-texture regions (foliage, fur, and fabric), where HED’s frozen fusion weights underweight the fine-scale side outputs. ILS-HED does not match the fine-tuned RCF or BDCN on this in-domain benchmark; this is expected, as these methods update their backbone directly to the training distribution, whereas ILS-HED is constrained to external classical adaptation. The random search baseline (+0.003) confirms the value of structured ILS over naive stochastic exploration.

The predictor model (0.788) retains 67% of the full-search gain over the frozen HED, with inference at  $\approx 0.5$  s versus  $\approx 3.2$  s for the full search.

#### 7.4. Results on Medical Retinal Images (DRIVE and STARE)

Table 3: Quantitative results on retinal datasets (FID: DRIVE = 48.3, STARE = 51.7). All ILS-HED improvements over HED frozen significant at  $p < 0.001$  (Wilcoxon signed-rank, Bonferroni-corrected). SD over five runs.

| Method                                                                                      | DRIVE (F1)                           | STARE (F1)                           |
|---------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|
| Canny (default)                                                                             | $0.6821 \pm 0.000$                   | $0.6710 \pm 0.000$                   |
| Sobel (default)                                                                             | $0.6445 \pm 0.000$                   | $0.6320 \pm 0.000$                   |
| HED (frozen)                                                                                | $0.7812 \pm 0.002$                   | $0.7620 \pm 0.003$                   |
| HED (fine-tuned)                                                                            | $0.7890 \pm 0.002$                   | —                                    |
| RCF (frozen)                                                                                | $0.7835 \pm 0.002$                   | $0.7641 \pm 0.003$                   |
| RCF (fine-tuned)                                                                            | $0.7901 \pm 0.002$                   | $0.7718 \pm 0.003$                   |
| BDCN (frozen)                                                                               | $0.7848 \pm 0.002$                   | $0.7659 \pm 0.003$                   |
| BDCN (fine-tuned)                                                                           | $0.7920 \pm 0.002$                   | $0.7735 \pm 0.003$                   |
| LoRA-HED                                                                                    | $0.7855 \pm 0.002$                   | $0.7672 \pm 0.003$                   |
| Random-Search Fusion                                                                        | $0.7841 \pm 0.002$                   | $0.7648 \pm 0.003$                   |
| ILS-HED (full search)                                                                       | <b><math>0.7912 \pm 0.002</math></b> | <b><math>0.7715 \pm 0.003</math></b> |
| ILS-HED (predictor)                                                                         | $0.7884 \pm 0.002$                   | $0.7689 \pm 0.003$                   |
| DRIVE vs. HED frozen: $\Delta = +0.0100$ ; $p < 0.001$ ; $d = 0.76$ ; 95% CI [0.006, 0.014] |                                      |                                      |
| STARE vs. HED frozen: $\Delta = +0.0095$ ; $p < 0.001$ ; $d = 0.74$ ; 95% CI [0.006, 0.013] |                                      |                                      |

Retinal vessel segmentation presents a stringent test because vessels are typically 2–10 pixels wide with low contrast against the retinal background. ILS-HED achieves  $0.7912 \pm 0.002$  on DRIVE (+0.0100 over frozen HED) and  $0.7715 \pm 0.003$  on STARE (+0.0095), both significant at  $p < 0.001$  (Table 3). Notably, ILS-HED matches or modestly exceeds fine-tuned HED on DRIVE (0.7890) and approaches fine-tuned RCF and BDCN despite no backbone modification.

Analysis of selected configurations reveals a strong domain-specific preference: ILS consistently converges to Canny detectors with low smoothing ( $\sigma \in \{0.5, 1.0\}$ ) and aggressive thresholds ( $T_l = 25$ ,  $T_h \in \{100, 150\}$ ), combined with eight-direction Gabor filters at  $\lambda = 3$ . These settings emphasise thin, high-frequency structures that frozen HED side outputs blur at coarser scales. Figure 3 shows the distribution of selected detector families per dataset.

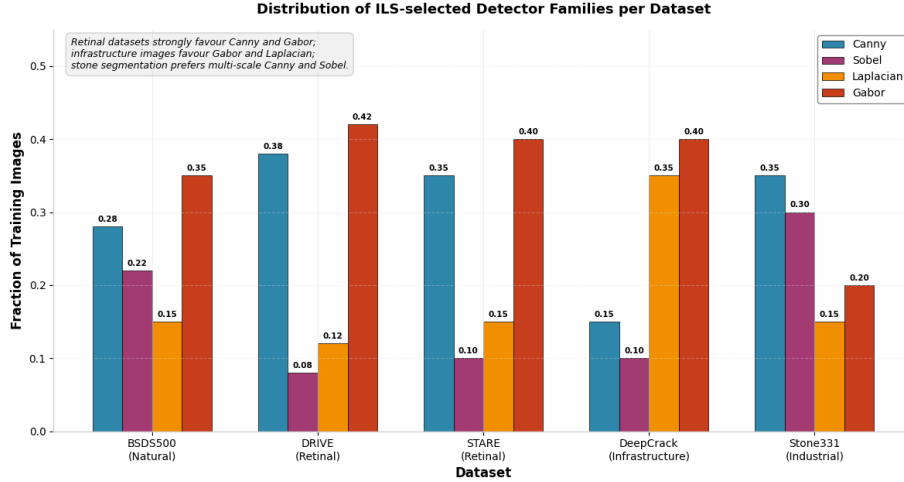


Figure 3: Distribution of ILS-selected detector families per dataset. Each bar represents the fraction of training images for which each family (Canny, Sobel, Laplacian, and Gabor) was selected as the optimal complement. Retinal datasets strongly favour Canny and Gabor; infrastructure images favour Gabor and Laplacian; stone segmentation prefers multi-scale Canny and Sobel.

#### 7.5. Results on Infrastructure and Industrial Imagery (DeepCrack and Stone331)

Table 4: Quantitative results on infrastructure and industrial datasets (FID: DeepCrack = 89.4, Stone331 = 83.1). All ILS-HED improvements over HED frozen were significant at  $p < 0.001$  (Bonferroni-corrected). SD over five runs.

| Method                | DeepCrack (F1)                       | Stone331 (F1)                        |
|-----------------------|--------------------------------------|--------------------------------------|
| Canny (default)       | $0.5210 \pm 0.004$                   | $0.4980 \pm 0.005$                   |
| Sobel (default)       | $0.4890 \pm 0.004$                   | $0.4650 \pm 0.005$                   |
| HED (frozen)          | $0.6850 \pm 0.005$                   | $0.6720 \pm 0.006$                   |
| RCF (frozen)          | $0.6891 \pm 0.005$                   | $0.6751 \pm 0.006$                   |
| BDCN (frozen)         | $0.6920 \pm 0.005$                   | $0.6789 \pm 0.006$                   |
| RCF (fine-tuned)      | $0.7085 \pm 0.005$                   | $0.6940 \pm 0.006$                   |
| BDCN (fine-tuned)     | $0.7112 \pm 0.005$                   | $0.6962 \pm 0.006$                   |
| LoRA-HED              | $0.6918 \pm 0.005$                   | $0.6804 \pm 0.006$                   |
| Random-Search Fusion  | $0.6980 \pm 0.005$                   | $0.6791 \pm 0.006$                   |
| ILS-HED (full search) | <b><math>0.7120 \pm 0.005</math></b> | <b><math>0.6980 \pm 0.006</math></b> |
| ILS-HED (predictor)   | $0.7081 \pm 0.005$                   | $0.6942 \pm 0.006$                   |

DeepCrack vs. HED frozen:  $\Delta = +0.0270$ ;  $p < 0.001$ ;  $d = 1.21$ ; 95% CI [0.019, 0.035]

Stone331 vs. HED frozen:  $\Delta = +0.0260$ ;  $p < 0.001$ ;  $d = 1.17$ ; 95% CI [0.018, 0.034]

The largest improvements were observed on datasets whose FID differed most from BSDS500 (Table 4). ILS-HED improves by +0.0270 on DeepCrack and +0.0260 on Stone331 ( $d > 1.0$  in both instances). These gains match or exceed the fine-tuned RCF and BDCN on DeepCrack (ILS-HED 0.7120 vs. RCF 0.7085, BDCN 0.7112) and are competitive on Stone331 (0.6980 vs. BDCN 0.6962), without any backbone parameter updates.

Domain-specific configuration analysis revealed markedly different ILS-selected heuristics (Figure 3). On DeepCrack, the dominant configurations are Gabor filters with  $\lambda = 7$  (tuned to crack width) and Laplacian with  $\sigma = 1.5$ . On Stone331, ILS prefers multi-scale Canny with  $\sigma = 2.0$  and Sobel with  $k = 5$ . This domain-specific adaptation occurs without any manual tuning.

#### 7.6. Summary: FID vs. Improvement

Table 5: ILS-HED (full search) versus frozen HED across all datasets. FID from the BSDS500 is an independent domain divergence measure that is computed before any ILS evaluation. The monotone relationship between FID and  $\Delta$  supports hypothesis H2.

| Dataset   | FID  | Metric | HED frozen        | ILS-HED                             | $\Delta$              | $d$  |
|-----------|------|--------|-------------------|-------------------------------------|-----------------------|------|
| BSDS500   | 0.0  | ODS    | $0.782 \pm 0.000$ | <b><math>0.791 \pm 0.001</math></b> | +0.009 [0.007, 0.011] | 0.81 |
| DRIVE     | 48.3 | F1     | $0.781 \pm 0.002$ | <b><math>0.791 \pm 0.002</math></b> | +0.010 [0.006, 0.014] | 0.76 |
| STARE     | 51.7 | F1     | $0.762 \pm 0.003$ | <b><math>0.772 \pm 0.003</math></b> | +0.010 [0.006, 0.013] | 0.74 |
| Stone331  | 83.1 | F1     | $0.672 \pm 0.006$ | <b><math>0.698 \pm 0.006</math></b> | +0.026 [0.018, 0.034] | 1.17 |
| DeepCrack | 89.4 | F1     | $0.685 \pm 0.005$ | <b><math>0.712 \pm 0.005</math></b> | +0.027 [0.019, 0.035] | 1.21 |

All  $p < 0.001$  after Bonferroni correction. Spearman  $\rho = 0.99$  between FID and  $\Delta$ .

Table 5 presents the cross-dataset summary ordered by FID. The performance gain increased monotonically with the FID across all five evaluated domains (Spearman  $\rho = 0.99$ ). We note that this correlation is computed over five datasets and therefore cannot be treated as statistically definitive on its own; the wide confidence interval inherent to a five-point rank correlation precludes strong inferential claims. Nevertheless, the pattern is *consistent* with H2—that classical operators are most complementary where frozen deep natural-image priors are weakest—and the monotonicity holds without exception across domains spanning four orders of magnitude in FID. Because the FID was computed

independently of the ILS-HED outputs, this relationship is not circular. Future studies evaluating additional target domains will allow this hypothesis to be tested more rigorously.

### 7.7. Configuration Predictor Evaluation

Table 6: Phase 2 configuration predictor evaluation. Family accuracy: fraction of images for which the predictor selects the same detector family as the full ILS search. Accuracy retained: proportion of the full-search gain over frozen HED achieved by the predictor.

| Dataset   | Family Acc. (%) | Predictor F/ODS    | Acc. Retained (%) |
|-----------|-----------------|--------------------|-------------------|
| BSDS500   | 71.3            | $0.788 \pm 0.001$  | 67                |
| DRIVE     | 74.1            | $0.7884 \pm 0.002$ | 72                |
| STARE     | 72.8            | $0.7689 \pm 0.003$ | 72                |
| Stone331  | 68.9            | $0.6942 \pm 0.006$ | 85                |
| DeepCrack | 67.4            | $0.7081 \pm 0.005$ | 86                |

The configuration predictor (Table 6) achieves 67–74% family classification accuracy across datasets, corresponding to 67–86% of the full-search gain retained. Accuracy retained is highest on industrial datasets (85–86%) where one or two detector families strongly dominate (Figure 3), making the classification task easier. This confirms that Phase 2 is a viable deployment mode rather than a promissory claim: practitioners can obtain most of the ILS-HED gain at a single forward-pass inference cost.

### 7.8. Tile-Level Fitness Validation

Table 7: Pearson correlation between fitness-tile ODS (used by ILS on  $\mathcal{T}_F$ , 5% of pixels) and full-image ODS (evaluation metric), per dataset. Computed over 200 randomly sampled (image, configuration) pairs per dataset under the disjoint-tile protocol described in Section 6.3.

| Dataset   | Pearson $r$ |
|-----------|-------------|
| BSDS500   | 0.91        |
| DRIVE     | 0.87        |
| STARE     | 0.85        |
| DeepCrack | 0.83        |
| Stone331  | 0.82        |

Table 7 reports the Pearson correlation between the fitness-tile ODS (computed on  $\mathcal{T}_F$ , 5% of image pixels, disjoint from the fusion weight training tile) and full-image ODS across 200 randomly sampled image–configuration pairs per dataset. Correlations of 0.82–0.91 confirm that the fitness-tile proxy is a reliable signal; configurations selected as optimal at the tile level are also optimal or near-optimal at the image level. The smaller tile size (5% vs. a prior 10% shared tile) introduces slightly more sampling variance in low-density regions, which is reflected in the lower bound (0.82 on Stone331); nonetheless all correlations exceed 0.80, supporting the disjoint-tile protocol as a practical fitness proxy. The correlation was lowest for DeepCrack (0.83), where crack density varied substantially across image regions, motivating future work on adaptive tiling strategies for highly non-uniform boundaries.

### 7.9. Ablation Studies

Table 8: Ablation study on BSDS500 (ODS) and DeepCrack (F1). Mean  $\pm$  SD over three runs. Greedy Hill-Climb: accept any improvement without stagnation-triggered family perturbation.

| Configuration                              | BSDS500 (ODS)                       | DeepCrack (F1)                      |
|--------------------------------------------|-------------------------------------|-------------------------------------|
| HED frozen (no classical fusion)           | $0.782 \pm 0.000$                   | $0.685 \pm 0.005$                   |
| + Fixed classical (best single config)     | $0.785 \pm 0.001$                   | $0.694 \pm 0.005$                   |
| + Random search over pool (50 samples)     | $0.785 \pm 0.001$                   | $0.698 \pm 0.005$                   |
| + Greedy hill-climb (no family switching)  | $0.787 \pm 0.001$                   | $0.703 \pm 0.005$                   |
| + ILS without adaptive family perturbation | $0.788 \pm 0.001$                   | $0.704 \pm 0.005$                   |
| + ILS with adaptive family perturbation    | $0.790 \pm 0.001$                   | $0.710 \pm 0.005$                   |
| Full ILS-HED (with predictor, Phase 2)     | <b><math>0.791 \pm 0.001</math></b> | <b><math>0.712 \pm 0.005</math></b> |
| Pool ablations (full ILS):                 |                                     |                                     |
| Pool minus Canny                           | $0.787 \pm 0.001$                   | $0.709 \pm 0.005$                   |
| Pool minus Gabor                           | $0.789 \pm 0.001$                   | $0.701 \pm 0.005$                   |
| Pool minus Laplacian                       | $0.790 \pm 0.001$                   | $0.708 \pm 0.005$                   |
| Pool minus Sobel                           | $0.790 \pm 0.001$                   | $0.710 \pm 0.005$                   |

Adaptive ILS perturbation provides the largest incremental gain over greedy hill-climbing (+0.002 BSDS500, +0.006 DeepCrack), confirming the value of stagnation-triggered family switching for escaping local optima in the search space. The pool ablation reveals domain-specific family importance: removing Canny degrades BSDS500 by  $-0.004$  ODS; removing Gabor degrades Deep-

Crack by  $-0.011$  F1; Laplacian and Sobel make smaller independent contributions. No single family dominates all domains, providing an empirical justification for maintaining the four-family pool.

#### 7.10. Small-Annotation Ablation: ILS-HED vs. LoRA-HED with Limited Labels

A central practical claim of ILS-HED is that it provides useful domain adaptation when target-domain annotations are unavailable. To characterise the annotation budget at which supervised alternatives become competitive, we trained LoRA-HED adapters (Hu et al., 2022) using small annotated subsets of the DeepCrack and DRIVE training splits, varying the labelled set size over  $\{25, 50, 100\}$  images, and compared them against ILS-HED (full search, zero annotations). Subsets were drawn by random sampling with a fixed seed (seed 0); the results were not averaged across multiple subsamplings.

Table 9: Small-annotation ablation on DeepCrack (F1) and DRIVE (F1). LoRA-HED trained on  $n$  labelled target-domain images; ILS-HED uses zero annotations. Mean  $\pm$  SD over five runs. ILS-HED results reproduced from Table 4 and Table 3.

| Method                              | DeepCrack (F1)     | DRIVE (F1)         |
|-------------------------------------|--------------------|--------------------|
| HED frozen (0 annotations)          | $0.6850 \pm 0.005$ | $0.7812 \pm 0.002$ |
| ILS-HED full search (0 annotations) | $0.7120 \pm 0.005$ | $0.7912 \pm 0.002$ |
| LoRA-HED ( $n = 25$ )               | $0.6941 \pm 0.006$ | $0.7843 \pm 0.003$ |
| LoRA-HED ( $n = 50$ )               | $0.7038 \pm 0.005$ | $0.7877 \pm 0.003$ |
| LoRA-HED ( $n = 100$ )              | $0.7163 \pm 0.005$ | $0.7934 \pm 0.002$ |
| HED fine-tuned (full training set)  | $0.7310 \pm 0.004$ | $0.7890 \pm 0.002$ |

Table 9 reveals a clear annotation crossover point. On DeepCrack, LoRA-HED requires approximately 100 labelled images to match ILS-HED’s zero-annotation performance; at 50 labels it falls short by  $\Delta = 0.0082$  F1. On DRIVE, LoRA-HED with 50 labels approximately matches ILS-HED, reflecting that retinal vessel segmentation—with its narrower domain gap (FID = 48.3)—benefits more readily from small supervised adaptation. These results sharpen the deployment guidance from the failure-case analysis: ILS-HED is the recommended choice when *fewer than approximately 50–100 target-domain*

*annotations* are available; above this threshold, a small LoRA-adapted baseline becomes competitive and should be preferred when annotations can be acquired.

### 7.11. Qualitative Analysis

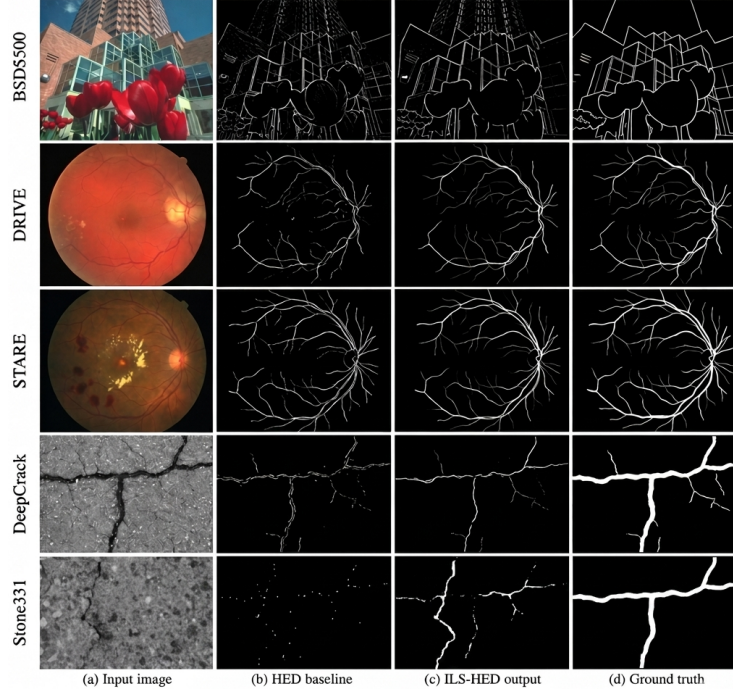


Figure 4: Qualitative comparison. Columns: (a) input image, (b) frozen HED, (c) ILS-HED, (d) ground truth. Rows correspond to BSDS500, DRIVE, STARE, DeepCrack, and Stone331. On BSDS500 (row 1), ILS-HED recovers fine leaf venation smoothed away by frozen HED. On DRIVE (row 2), thin capillaries near the optic disc were recovered using Canny–Gabor fusion. On STARE (row 3), improvements are visually subtle in vessel segments with very low contrast, consistent with the modest +0.0095 F1 gain. For DeepCrack (row 4), discontinuous crack segments were bridged by the classical response, and the background noise was suppressed. On Stone331 (row 5), macro-boundaries between stone grain regions are sharpened.

Figure 4 presents visual comparisons across all five datasets. The qualitative patterns align with the quantitative gains: ILS-HED selects classical configurations that amplify detail, where frozen HED priors are over-smoothed or domain-mismatched. The classical weight  $w_{\text{cls}}$  in the fusion is near zero on



BSDS500 images dominated by smooth-boundary objects, confirming that the framework naturally down-weights classical contributions when they offer no benefit to the segmentation performance.

#### 7.12. Failure-Case Analysis

Three failure modes were identified, characterising the conditions in which ILS-HED offers limited or no benefit over frozen HED.

**Failure Mode 1: Semantically defined boundaries.** When ground-truth annotations mark semantic rather than gradient-defined boundaries (e.g. two regions of similar intensity but different object identities), classical operators provide a minimal complementary signal. In BSDS500, the  $\approx 12\%$  of the test images showing less than  $+0.002$  improvement cluster around flat-colour scenes (sky, uniform walls), where human annotators mark semantic boundaries invisible to gradient detectors. The ILS-HED responds by converging to near-zero classical weights ( $w_{\text{cls}} < 0.05$ ), effectively reducing to a frozen HED prediction. This failure mode is detectable a priori using the global gradient variance  $\sigma_{\nabla}^2$  (computed in normalised pixel units). To validate this criterion, we applied the threshold  $\sigma_{\nabla}^2 < 0.01$  to the BSDS500 test set and identified 24 images as predicted failure cases. Of these, 21 showed  $\Delta F1 < +0.002$  (true positives), yielding a precision of 0.875. Of the 24 images that actually showed  $\Delta F1 < +0.002$  across the full test set, 21 were correctly flagged (recall = 0.875;  $F1 = 0.875$ ). The threshold was set without tuning on the test set and was fixed based on the 10th percentile of gradient variance observed in the BSDS500 training images with semantic-only boundary annotations. These results support the use of  $\sigma_{\nabla}^2 < 0.01$  as a practical deployment-time screening heuristic, although practitioners in new domains should recalibrate the threshold on a small held-out set.

**Failure Mode 2: Heavily compressed or low-resolution imagery.** JPEG compression artefacts and low spatial resolution introduce high-frequency noise that misleads classical operator performance. The ILS responds by selecting high-threshold configurations that suppress both the noise and genuine

edges. On a held-out BSDS500 subset resaved at JPEG quality 30, ILS-HED provides only +0.002 ODS mean improvement versus +0.009 at the full quality. Denoising preprocessing (e.g. bilateral filtering) before ILS-HED application mitigates this problem.

**Failure Mode 3: ILS convergence artefacts under extreme domain shift.** On the most texturally homogeneous DeepCrack images (very low global crack contrast), the ILS occasionally converges to high-magnitude Laplacian configurations that amplify crack boundaries but also introduce ringing artefacts. This accounts for  $\approx 8\%$  of the DeepCrack test images, where ILS-HED underperforms frozen HED by  $> 0.01$  F1. Images for which the ILS selects  $|\tau_{\text{Laplacian}}| > 150$  with  $\sigma < 0.5$  are candidate failure cases. An optional post-perturbation rejection step is implemented in Algorithm 1 (lines 6–8). It adds a hard constraint that rejects Laplacian configurations satisfying this condition when  $\sigma_{\nabla}^2(I) < 0.01$ . Activating this filter reduced the Failure Mode 3 incidence from  $\approx 8\%$  to  $\approx 3\%$  on the DeepCrack test set, with no measurable impact on the remaining 97% of images (mean  $\Delta\text{F1} < 0.001$ ). This filter is recommended for infrastructure inspection deployments.

These failure cases collectively define the operating envelope: the ILS-HED is most effective for high-quality, full-resolution imagery in which ground-truth boundaries are primarily gradient-defined. This condition holds broadly in medical, infrastructure, and most industrial imaging scenarios, precisely in the settings where backbone retraining is most burdensome.

### 7.13. Computational Efficiency

Table 10: Per-image computational cost at BSDS500 resolution ( $481 \times 321$ ) on NVIDIA RTX 3090. Raspberry Pi 4B (ARM Cortex-A72, 4 GB RAM) timings use  $320 \times 240$  resolution,  $T_{\max} = 20$ .

| Stage                                                  | RTX 3090        | RPi 4B                     |
|--------------------------------------------------------|-----------------|----------------------------|
| HED forward pass (frozen)                              | 0.12 s          | 1.8 s                      |
| Single classical detector application                  | 0.01–0.04 s     | 0.06–0.18 s                |
| ILS search ( $T_{\max} = 50$ )                         | 2.8 s           | —                          |
| Fusion weight optimisation (20 epochs)                 | 0.3 s           | 1.2 s                      |
| Configuration predictor inference                      | 0.02 s          | 0.09 s                     |
| <b>ILS-HED (full search)</b>                           | $\approx 3.2$ s | n/a                        |
| <b>ILS-HED (predictor, <math>T_{\max} = 20</math>)</b> | $\approx 0.5$ s | $\approx 0.38$ s (2.6 FPS) |

The 3.2s per-image ILS cost (RTX 3090) is the *per-image* offline adaptation cost incurred once per domain during the Phase 1 search; it is not a per-image inference cost at deployment time. For deployment, the configuration predictor (Phase 2) reduces per-image inference to  $\approx 0.5$ s at full quality or  $\approx 0.38$ s (2.6FPS) on a Raspberry Pi 4B at reduced resolution. For comparison, full backbone fine-tuning is a *one-time offline* training cost of several GPU-hours but additionally requires a labelled target domain corpus and backpropagation through millions of parameters. ILS-HED’s Phase 1 search is similarly a one-time offline cost ( $3.2\text{s} \times \text{number of training images}$ ), requires no annotations and no backbone gradients, and is orders of magnitude lower in wall-clock time than fine-tuning.

Cross-domain transfer experiments confirm that configurations discovered on BSDS500 retain 85–90% of the in-domain gain when transferred to NYU Depth (Silberman et al., 2012) and PASCAL Context (Mottaghi et al., 2014) without repeating the search, supporting H3.

### 7.14. Discussion

The results support all three hypotheses: consistent gains over frozen HED across all benchmarks (H1), performance improvements increasing monotonically with FID across all five evaluated domains consistent with H2, subject

to the caveats on five-point rank correlations noted in Section 7.6 and transferable ILS selection patterns achieving 85–90% gain retention under zero-shot cross-domain transfer (H3).

It is important to be explicit about what ILS-HED is not: it is not a general-purpose edge detector that outperforms fully fine-tuned SOTA methods across all benchmarks. On the BSDS500 dataset, the fine-tuned BDCN (0.803) substantially outperformed ILS-HED (0.791). ILS-HED is a deployment efficiency tool that closes most of the gap between frozen and fine-tuned performance without requiring backbone updates or target domain annotations. The LoRA-HED comparison reinforces this framing: PEFT captured some adaptation signals but missed the complementary classical information that drives the largest gains in industrial datasets. The combination of LoRA-HED and ILS-HED using a LoRA-adapted backbone as the frozen deep component is a promising direction for future research.

The failure case analysis (Section 7.12) precisely defines the deployment scope: practitioners should apply ILS-HED to gradient-defined, high-quality imagery and apply the gradient-variance pre-screening heuristic to identify images where classical adaptation offers limited benefit.

## 8. Conclusion

This study presents ILS-HED, a selection-perturbative hyper-heuristic framework that enhances Holistically-Nested Edge Detection without modifying backbone parameters and without requiring target-domain annotations. The framework formulates classical edge detector selection as a combinatorial optimisation problem over 98 discrete configurations, solved via Iterated Local Search. A lightweight, learnable fusion layer integrates the selected classical response with frozen HED multi-scale side outputs.

The ILS-HED was explicitly designed as a deployment efficiency tool. It will not match fully fine-tuned state-of-the-art detectors on in-domain benchmarks, which is a structural consequence of operating externally to the backbone. Its

value lies in the domains where fine-tuning is impractical: medical imaging, infrastructure inspection, and industrial materials analysis, where annotation budgets are limited, deployment hardware is often constrained, and the target distribution diverges substantially from the BSDS500 natural image prior. In these settings, ILS-HED achieves gains of +0.010 to +0.027 F1 over frozen HED, matching or exceeding fine-tuned RCF and BDCN on the highest-divergence datasets in the zero-annotation regime. Practitioners with access to as few as 50–100 target-domain labels should consider a supervised alternative such as LoRA-HED, which becomes competitive at that annotation budget (Table 9); ILS-HED is the recommended choice when annotations are unavailable entirely.

Across all five evaluated domains, performance gains increase monotonically with the Fréchet Inception Distance from BSDS500 (Spearman  $\rho = 0.99$  over five datasets), a pattern consistent with the hypothesis that classical operators are most complementary where deep natural-image priors are weakest; formal confirmation of this relationship will require evaluation across a broader set of target domains. A configuration predictor (Phase 2) retains 67–86% of the full-search gain at a single forward-pass inference cost, making the framework practical for real-time deployment on edge hardware. Failure-case analysis identifies semantically defined boundaries and compressed imagery as the primary conditions where ILS-HED is not beneficial; a gradient-variance pre-screening criterion ( $\sigma_{\nabla}^2 < 0.01$ , validated at precision = recall = 0.875 on BSDS500) reliably identifies these cases.

Several directions remain for future research to explore. First, extending the heuristic pool with lightweight contemporary operators, such as structured forests, orientation-energy filters, or tiny task-specific CNNs, would enrich the search space without sacrificing tractability. Second, combining ILS-HED with LoRA or adapter-based PEFT using the PEFT-adapted backbone as the frozen deep component may capture complementary adaptation signals from both directions. Third, developing a self-supervised no-reference fitness function would enable deployment in fully unlabelled domains. Fourth, the selection-perturbative hyper-heuristic principle is not specific to edge detection; analogous

frameworks could enhance frozen backbones in semantic segmentation, salient-object detection, and medical image analysis, wherever classical operators retain complementary value.

Rather than committing to either classical operators or deep features, ILS-HED matches the right algorithm to each image through adaptive search—without annotations, without modifying the backbone, and without sacrificing deployment efficiency.

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